CSE 318 — Online 4

Section B2

Plain Nearest Neighbor is short-sighted: it may pick a slightly cheaper next move that forces a very expensive move afterwards. A cheap fix is to look *one step further* when deciding.

What to implement.
Write `lookaheadNN()`, a modified nearest-neighbor construction. Let c denote the current city and U the set of unvisited cities.

1. If $|U| = 1$: move to the only remaining city and stop.
2. Otherwise, for each candidate $d \in U$, compute a lookahead score:

$$\text{score}(d) \;=\; c(c, d) \;+\; \min_{e \in U \setminus \{d\}} c(d, e).$$

   That is: the cost of going to d, plus the cost of the best next hop from d.
3. Pick the d with the **smallest** score. Ties: break by the immediate cost $c(c, d)$.
4. Move to d, mark it visited, repeat.
5. After all cities are visited, close the tour by returning to city 0.

Output:

- The lookahead tour and its cost.
- The plain 1-step Nearest Neighbor tour and cost on the same instance.
- Percentage improvement.
- Test on **at least three** instances of different sizes.